



Numerical Solution of Laplace's Equation

A Finite Difference Approach for a Rectangular Coaxial Cable

Abstract

This report details a numerical methodology for solving Laplace's equation in two dimensions. The primary objective is to determine the electric potential distribution, $V(x, y)$, within a rectangular coaxial cable utilizing the finite difference method (Jacobi iteration). This document presents the mathematical formulation, boundary conditions, and the corresponding computational implementation.

1 Problem Description

The physical system consists of a coaxial cable modeled in a two-dimensional Cartesian domain. The outer boundary is a rectangular region with dimensions $a \times b$. Inside this region lies a solid inner rectangular conductor with dimensions $c \times d$, positioned with an offset of (x_0, y_0) from the origin.

The boundary conditions governing the potential are defined as follows:

- The outer rectangular shell is grounded, enforcing a Dirichlet boundary condition of $V = 0$.
- The inner rectangular conductor is maintained at a constant high potential, V_0 .

2 Mathematical Formulation

In the charge-free dielectric region between the conductors, the electric potential satisfies Laplace's equation:

$$\nabla^2 V = 0 \quad (1)$$

In a 2D Cartesian coordinate system, this expands to:

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0 \quad (2)$$

To solve this numerically, we apply the central finite difference approximation to discretize the second-order partial derivatives over a uniform grid with spacing Δx and Δy :

$$\frac{V(x_{i+1}, y_j) - 2V(x_i, y_j) + V(x_{i-1}, y_j))}{(\Delta x)^2} + \frac{V(x_i, y_{j+1}) - 2V(x_i, y_j) + V(x_i, y_{j-1}))}{(\Delta y)^2} = 0 \quad (3)$$

Assuming a uniform resolution where $\Delta x = \Delta y$, the equation simplifies significantly. Solving for the potential at the central grid point (i, j) yields:

$$V(x_i, y_j) = \frac{1}{4} (V(x_{i+1}, y_j) + V(x_{i-1}, y_j) + V(x_i, y_{j+1}) + V(x_i, y_{j-1})) \quad (4)$$

This relationship forms the basis of the **Jacobi Iteration Method**. By repeatedly updating each grid point to be the arithmetic mean of its four adjacent neighbors, the numerical grid asymptotically converges toward the true physical solution.



3 Computational Implementation

A computational grid of $nx \times ny$ nodes is established across the domain. The iterative solver applies the Jacobi method for a designated number of iterations ($n = 1000$).

A critical aspect of the implementation is enforcing the boundary conditions at each iteration step. While the outer boundary naturally remains at $V = 0$ by avoiding updates to the grid edges, the inner conductor's nodes must be explicitly reset to V_0 after every iteration to prevent the potential from numerically diffusing.

3.1 Core Computational Logic

The following MATLAB code snippet demonstrates the vectorized implementation of the Jacobi iteration and the enforcement of the inner conductor's Dirichlet boundary condition:

```
1 % --- Iterative Numerical Solution (Jacobi Method) ---
2 iterative_number = 1000;
3
4 for iter = 1:iterative_number
5 % Update potential for all interior points using vectorization
6 V(2:end-1, 2:end-1) = 0.25 * (V(3:end, 2:end-1) + V(1:end-2, 2:end-1) + ...
7 V(2:end-1, 3:end) + V(2:end-1, 1:end-2));
8
9 % Re-enforce the fixed potential of the inner conductor
10 V(x_start:x_end, y_start:y_end) = V0;
11 end
```

Listing 1: Iterative Numerical Solution

The complete MATLAB script, including grid initialization and data visualization (2D contours and 3D surface plots), is provided in the GitHub repository.